Hard Shape Constraints versus Penalties for Temperature Extrapolation of MOF and COF Water Isotherms

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Abstract

Learned models of water isotherms in metal-organic and covalent organic frameworks (MOFs and COFs) can predict uptake that falls as pressure rises; monotonicity can be built into a model or added as a training penalty. In a pre-registered benchmark on 13 frameworks, models predicted the highest or lowest measured temperature, 5 to 25 K beyond the training data, and a model built to be monotone and linear at low pressure, P, was compared with a relaxed version of it trained with penalties, with neural networks and with classical models. The penalty version had falling or negative uptake in 9 of 26 extrapolations at its accuracy-tuned weight (none above the counting tolerance in a post hoc refit with far heavier weights), and the accuracy comparisons were inconclusive (error ratio to the penalty version 1.10, 95% confidence interval 0.93 to 1.65; non-inferiority within 25% not shown), and characteristic-curve models had the lowest median errors. P's heats were free in sign, so its isosteric heat left 0 to 150 $\mathrm{kJ\,mol^{-1}}$ in 14 extrapolations and its worst error came from this freedom:

shape constraints guarantee monotone isotherms without tuning, but the temperature dependence needs constraints of its own.

Introduction

Water sorbents built from metal-organic frameworks (MOFs) and covalent organic frameworks (COFs) are studied for atmospheric water harvesting[1,2] and for adsorption heat pumps and chillers.[3–5] These devices run between temperatures: a MOF-801 cooling cycle adsorbs at 30 °C and regenerates at 80 to 85 °C.[6] Its design needs the uptake at temperatures where no isotherm was measured, and the isosteric heat, which enters the cycle model.[5] Both rest on few temperatures. A practical guide asks for isotherms spaced by 20 K or less for a Clausius-Clapeyron heat and advises against extrapolation beyond the measured range.[7] For Mg-SU-102, isotherms were measured from 15 to 55 °C, but only those from 25 to 40 °C had enough points for the heat.[8]

The standard temperature model for water in MOFs is the characteristic curve: the uptake is written as a function of the adsorption potential $A = RT \ln(p_{\mathrm{sat}}/p)$, the isotherms of all temperatures are expected to fall on one curve, and that curve is used to model cycles.[9] Within this approach the isosteric heat is the vaporisation enthalpy plus A.[5] Invariance is usually shown by overlaying curves, without an error number, as for MIL-160 from 20 to 50 °C[1] and Al-fumarate from 298 to 333 K.[10] The Mg-SU-102 curves shift slightly towards lower energies as the temperature rises,[8] and MOF-303 loses a large part of its capacity between 25 and 45 °C.[11] Statistical thermodynamics gives a characteristic curve when added sorbate contributes purely energetically, and a temperature-dependent isotherm when sorbate number correlates with entropy.[12] The other common route, Langmuir or Sips sites with temperature-dependent affinities fitted to all temperatures at once,[5,13] has its own defects. Sips and Freundlich terms do not reduce to Henry's law at zero pressure (their slope there is zero or infinite) and their heats diverge at low coverage, dual-site Langmuir heats can show

an unphysical maximum, and models that fit equally well disagree on the heat wherever they extrapolate.[13]

Machine-learned isotherm models add flexibility but little thermodynamics. In two recent examples temperature is a plain input: one network predicted methane isotherms of 17,644 MOFs from 278 to 338 K and was extrapolated to 268 and 358 K;[14] another predicts water isotherms from pressure and structural properties.[15] Trained on molecular simulations, a deep network predicts binary sorption loadings in zeolites as a function of thermodynamic state variables.[16] Where heats appear, they are separate outputs. A neural ordinary differential equation learns $\mathrm{d}q/\mathrm{d}\ln p$ at one temperature, observes monotonicity without imposing it, and predicts the heat with a second network.[17] A zeolite study joined a learned loading at 300 K and an independently learned heat through an integrated Clausius-Clapeyron step with a heat that depends on loading but not on temperature.[18] When physics enters, it enters as a penalty. A network for hydrogen sorption adds monotonicity as a weighted loss term and reports 98.6% monotonicity satisfaction, so 1.4% of held-out cases violate it.[19] A penalty-trained network is not provably monotone; monotonicity can only be certified afterwards,[20] unless the architecture builds it in.[21]

A different construction is possible. The grand (immersion) potential of the adsorbed phase, $\Omega = -RT \int n \,\mathrm{d}\ln p$, gives the isotherm through its derivative with respect to $\ln p$, $n = -(\partial\Omega/\partial \ln p)/RT$.[22,23] A model can be written for this one scalar as a function of $\ln p$ and T. If it is convex in $\ln p$, the isotherm is monotone for every parameter value; input-convex networks give convexity by construction,[24] and so does a positive mixture of Langmuir and cooperative terms written as the derivative of a sum of softplus functions. A potential linear in p at low pressure gives a finite, non-zero Henry slope, which a Sips term with an exponent other than one lacks.[13] The isosteric heat then follows by Clausius-Clapeyron from second derivatives of the same function and is finite wherever $\mathrm{d}q/\mathrm{d}\ln p$ is positive. Neighbouring fields already build thermodynamic structure into learned models this way. A learned free energy gives fluid properties as its derivatives, with exact Maxwell relations.[25] In mixture

thermodynamics, a graph network for the excess Gibbs energy replaced a Gibbs-Duhem penalty[26] and needs no extra loss terms,[27] and hard-constraint networks build the Gibbs-Duhem relation into the architecture[28] and take the excess enthalpy from the temperature derivative of a learned excess Gibbs energy.[29] In adsorption, a learned density functional has produced an isotherm of a Lennard-Jones fluid in a model pore.[30] In the sources we searched we found no test of hard against penalty constraints for the temperature extrapolation of measured water isotherms.

Hard constraints make a model admissible, but not necessarily accurate. The question is older than chemistry: hard output constraints on deep networks proved computationally feasible but no better than soft ones,[31] while networks trained with the residual of a differential equation as a penalty[32] can face an ill-conditioned optimisation and fail to learn the physics.[33] For phase-transition enthalpies, the error of the predicted sublimation enthalpy rose by more than 10 $\mathrm{kJ\,mol^{-1}}$ under a hard thermodynamic constraint, while a soft constraint kept the accuracy.[34] Two parts of the construction can also act separately: the shape constraint fixes monotonicity and the Henry limit, while the temperature structure (a van't Hoff affinity per term, a free dependence on $1/T$, or a characteristic curve) decides how the isotherm moves between temperatures. The open question is which of the two, if either, makes extrapolation in temperature better than penalties, generic networks and classical models. A null answer would be as useful as a positive one, provided it can be told apart from an inconclusive one. Hard constraints restrict the architecture and raise the training loss; if they buy admissible isotherms at no accuracy cost, a modeller can adopt them for the guarantee alone, and if a classical model extrapolates as well, a learned model has to justify itself by something other than accuracy. A test that can say this needs a margin, fixed in advance, within which two models count as equivalent.

The present study therefore asks whether hard shape constraints make the temperature extrapolation of measured water isotherms better than penalties, generic networks and classical models. It does so in a pre-registered factorial design that crosses the shape constraint

(none, penalty or hard) with the temperature structure (free in $1/T$, van't Hoff per term, or a temperature-invariant characteristic curve). The reference model, P, is the cell with hard constraints and van't Hoff terms: a positive mixture of Langmuir and cooperative terms written as the derivative of one convex potential, fitted to each material. It is compared with its penalty and unconstrained twins, the hard-constrained cells of the other two temperature structures, a generic neural network and a dual-site Sips model with van't Hoff affinities. The data are 13 real sets of water isotherms measured at three or more temperatures, 5 on MOFs and 8 on COFs. Each model predicts a material's highest and lowest temperatures from the others, 5 to 25 K beyond the nearest training temperature. The design, tests and words of every verdict were fixed before any confirmatory fit, including words for a classical model that wins and for models equivalent within 25%. Every defect found in the design, code or data is recorded in a public correction ledger. None of the 13 materials has a calorimetric water heat; our search found one verified value for a MOF with multi-temperature isotherms, a single integral heat for MOF-801,[35] and none for any COF. We therefore test the heat on synthetic truths with exact heats, built on each material's own temperatures and pressures.

The study makes four contributions. (i) A benchmark protocol for constraint strategies in learned models of scientific data, applied here to measured water isotherms: a pre-registered factorial design, synthetic data on which each verdict is checked to be able to fail, the power at the margin, scoring by the frozen analysis code and by a second scorer written separately from the frozen text, and a public record of every deviation. (ii) An inconclusive accuracy comparison of the hard shape constraints with the penalty (error ratio 1.10, 0.93 to 1.65; no cost was detected, and one larger than the margin could not be excluded), and that they differ in how the shape is obtained: at the weight chosen for accuracy the penalty left shape violations in 9 of 26 extrapolations, and far heavier weights removed them in a post hoc refit, whereas the hard constraints exclude them by construction. (iii) The finding that the same model's heats left the physical range in 14 of 26 extrapolations, because

its temperature dependence is free in sign, which traces its worst failure. (iv) Descriptive evidence suggesting that the temperature structure, more than the shape constraints, limits temperature extrapolation and the heat.

Results

The design and the run

The models form a factorial of shape constraint (none, penalty, hard) and temperature structure (free in $1/T$, a van't Hoff affinity per term, a characteristic curve), and P is its hard van't Hoff cell (Figure 1). Each material is fitted in three tasks: E-high and E-low hold out the highest or the lowest temperature, and I-mid holds out a middle one. The confirmatory run produced 432 fits: 11 cells on 13 materials and three tasks, and the Kirchhoff variant P-K on the one material with at least four training temperatures, as pre-registered. Every fit was started from 5 seeds, and zero fits failed. The pre-registered analysis and the second scorer, written separately from the frozen text alone, gave the same ratios, intervals, adjusted p values and admissibility classes. Table 1 lists the models and their labels.

Accuracy of temperature extrapolation (H1)

None of the four confirmatory comparisons separated P from its comparator (Figure 2a). Against the penalty model V-pen, a relaxation of P's function class (signed capacities, free exponents) with the shape conditions as a training penalty, the ratio of extrapolation errors was $R = 1.10$ (exact 95% interval 0.93 to 1.65; Holm-adjusted $p = 0.76$). The pre-registered claim, that the hard constraints cost less than 25% in accuracy, required the upper bound to fall below 1.25; it did not, so the claim is not supported (the upper 98.75% bound, which applies the family correction, is 3.08). Under equal accuracy the criterion would have been met with probability 0.51 to 0.74 at the spread of per-material differences assumed in the power analysis, 0.158 decades (Note S1); the observed spread was 0.30 decades, about

Table 1: Models and labels (definitions in Note S4).

Label	Shape constraint	Temperature structure	Role and form
P	hard	van't Hoff per term	reference: positive mixture of Langmuir and cooperative terms, derivative of one convex potential
V-pen	penalty	van't Hoff per term	confirmatory comparator (H1b): P's function with signed capacities and free exponents, monotonicity and Henry penalties
V-none	none	van't Hoff per term	descriptive: V-pen without penalties
A-hard	hard	characteristic curve	confirmatory comparator (H1c): uptake a decreasing function of the adsorption potential A
A-pen, A-none	penalty, none	characteristic curve	descriptive: A-hard with signed terms, with or without the monotonicity penalty
F-hard	hard	free in $1/T$	descriptive (H1a): P's potential with capacities and positions from a network of $1/T$
F-none, F-pen	none, penalty	free in $1/T$	confirmatory comparator (H1d, F-none): multilayer perceptron of $\ln p$ and $1/T$; F-pen adds the penalties
F-rel	none	free in $1/T$	descriptive: as F-none with $\ln(p/p_{\mathrm{sat}})$ as input
C1	monotone by construction	van't Hoff per site	confirmatory comparator (H1e): dual-site Sips
P-K	hard	van't Hoff with heat capacity	descriptive: P with a Kirchhoff term
P-sign	hard, signed heats	van't Hoff, $Q_k > 0$, $\alpha_k \geq 0$	exploratory, after the verdicts

twice that. The interval includes both a small advantage for P and a substantial one for V-pen. Against the hard characteristic curve A-hard, $R = 1.51$ (0.84 to 2.87); against the unconstrained network F-none, $R = 0.63$ (0.40 to 1.22); against dual-site Sips with van't Hoff affinities (C1), $R = 1.23$ (0.92 to 1.96). All four Holm-adjusted p values were 0.76, and each verdict reads "inconclusive at this sample size".

One material dominates the spread. On COF-SO_3H, a COF carrying sulfonic acid groups, P's extrapolation error was 10.9 times that of V-pen, 9.9 times that of A-hard, 6.2 times that of F-none and 8.6 times that of C1, the largest ratio in every comparison (Table S2). On the other materials the ratios scatter about one: P was more accurate than V-pen on 6 of the 13

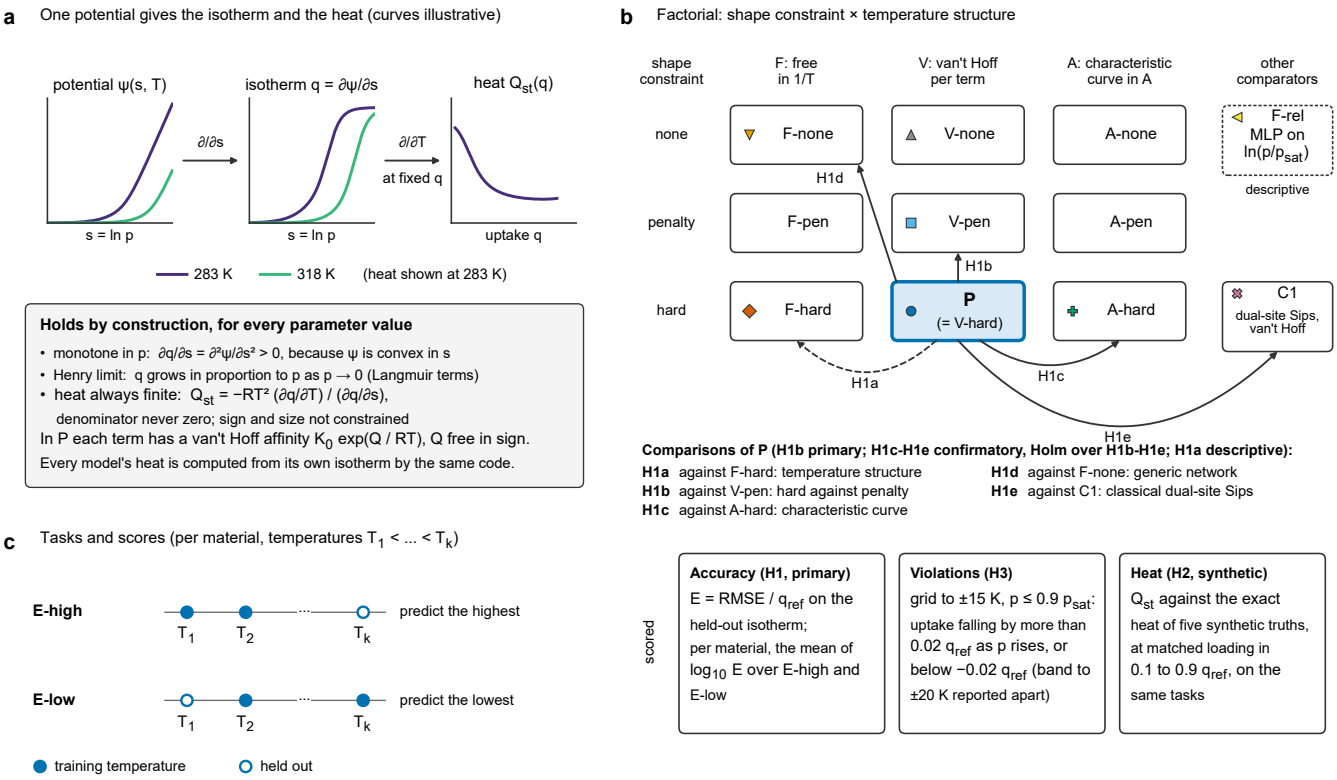


Figure 1: **Model, factorial and tasks.** **a**, The potential $\psi(s, T)$ of P gives the isotherm as its derivative in $s = \ln p$ and the heat from the isotherm; the three properties in the box hold for every parameter value. **b**, The factorial of shape constraint and temperature structure, the two further comparators and the comparisons of P with their evidence labels. **c**, The extrapolation tasks and the three scores.

materials, than F-none on 8, than C1 on 6 and than A-hard on 5. Split by class (estimates only, 5 MOFs and 8 COFs), the ratio against V-pen was similar for the two ($R = 1.09$ and 1.10), while the characteristic curve did much better than P on the COFs than on the MOFs ($R = 2.67$ against 0.69); the generic network and C1 were also relatively better on the COFs (Table S7).

The descriptive comparison with the free temperature network F-hard, which shares P's shape constraints but has no temperature law, favoured P ($R = 0.49$, 0.33 to 0.85, unadjusted $p = 0.027$; smaller error on 11 of 13 materials). The synthetic worlds showed that this comparison cannot fail when only two temperatures are available for training (Note S1), so it is reported as a description and not as a test of the temperature law. On the two materials

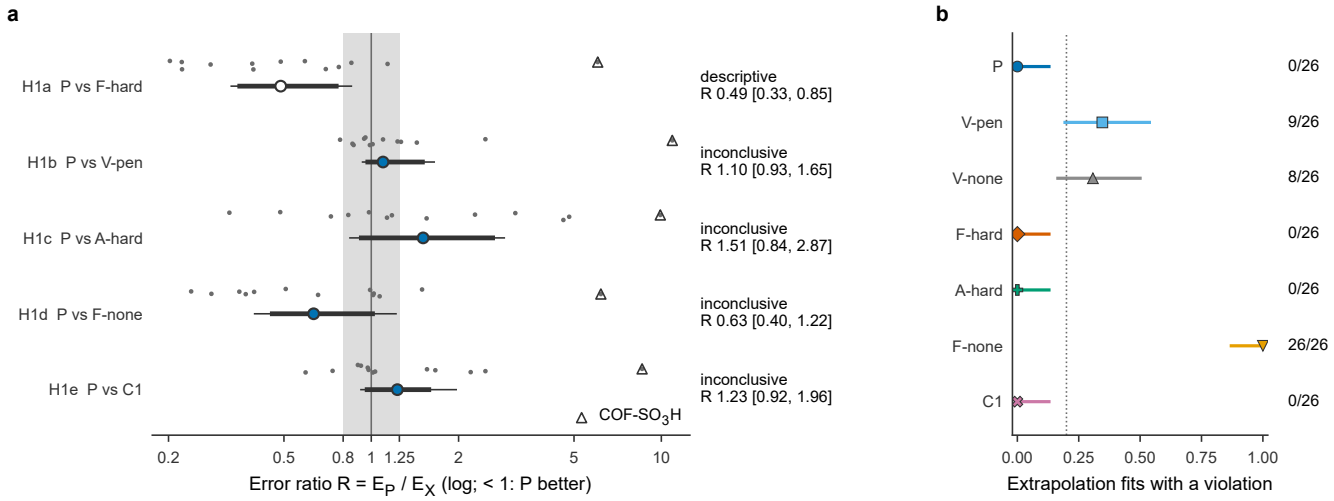


Figure 2: **Confirmatory verdicts. a**, Ratio of extrapolation errors $R = E_P/E_X$ (log scale; below one, P is more accurate): Hodges-Lehmann estimate with exact 95% (thin) and 90% (thick) intervals over the 13 materials; dots, single materials (open triangles, COF-SO_3H); shading, the equivalence band 0.8 to 1.25. H1a (open circle) is descriptive and not part of the confirmatory family. **b**, Fraction of the 26 extrapolation fits (E-high and E-low) with a violation inside the domain, with Wilson 95% intervals; dotted line, the pre-registered 20% threshold; the other models are in Table S3.

with at least three training temperatures, where the free network has more to learn from, P was also more accurate ($R_m = 0.28$ for Mg-SU-102 and 0.49 for Zn_2(bptc)).

Admissibility of the extrapolated isotherms (H3)

No extrapolation fit of P, F-hard, A-hard or C1 had a shape violation inside the domain (0 of 26 each; upper 95% bound 0.13; Figure 2b), and the counts of decreasing or negative uptake on the whole grid, which must be zero by construction, were zero for P, F-hard, A-hard and P-K (0 points), as required. These zeros follow from the construction of the four models (and of C1, a positive sum of increasing terms); the outcome of the test lies in the models without hard constraints. The penalty model V-pen violated in 9 of 26 fits (Wilson interval 0.19 to 0.54), no fewer than the unpenalised V-none (8 of 26). The pre-registered prediction, at least one violating fit for V-pen and none for P, was therefore met. Without the van't Hoff structure the penalty helped little: F-pen violated in 13 fits and A-pen in 16, against 26

for F-none and 15 for A-none, and the relative-pressure network F-rel violated in 22. With no tolerance and over the whole grid, V-pen had 11 violating fits (Table S3).

Because one weight multiplies both of V-pen's penalties and was chosen for accuracy, its violations might only mean that the monotonicity penalty was too weak. An exploratory refit specified after the verdicts (post hoc) tested this: V-pen was refitted on the confirmatory materials with the monotonicity penalty on its own weight, a fixed grid of 10^2, 10^4 and 10^6 times the tuned λ that was not tuned on the development materials, and everything else as in the confirmatory run. The violating fits fell from 9 to 4, 2 and 0 of 26, and at the heaviest weight none remained in the extrapolated band either (0), while the error ratio against P changed little ($R = 1.07$, 1.09 and 1.12 against 1.10 at the tuned weight, every 95% interval containing one; Table S9). Even at the heaviest weight 5 fits still had decreasing or negative points smaller than the tolerance of the count, 2 of them with the penalty exactly zero on its collocation points.

The shape constraints do not cover the heat's sign or size, and the pre-registered count of heats outside 0 to 150 $\mathrm{kJ\,mol^{-1}}$ shows it. The count is taken over the whole evaluation grid, which reaches 20 K beyond the measured temperatures; the penalty domain is the part within 15 K of them and up to 0.9 p_{sat}. P's heat was never undefined, but 14 of its 26 extrapolation fits had grid points outside that range (13 inside the penalty domain): 10 with negative heats (9 of them inside the penalty domain) and 6 with heats above 150 $\mathrm{kJ\,mol^{-1}}$. The free temperature network F-hard, which shares P's shape constraints, had 23 such fits. The penalty model and the same function without a penalty had similar counts (12 and 15 fits) and in addition undefined heats where their isotherms decreased (11 and 8 fits); C1 had 2 such fits and A-hard none, because its heat is the vaporisation enthalpy plus a positive adsorption potential. In P the heats Q_k and the expansion coefficients α_k are free in sign: 24 of its fits had a term whose capacity rises with temperature ($\alpha_k < 0$) and 6 a term with a negative heat (refits of the reported seeds, post hoc).

Isosteric heat on synthetic truths (H2)

The fifteen heat tests compare P with V-pen, A-hard and F-none on each truth family, sampled on the confirmatory materials' own temperatures and pressures, with no direction predicted (Figure 3 and Table S5). P's heat was never undefined: zero of its tasks failed, against 28 of the 130 tasks of the unconstrained network. Over the tasks with a defined heat, the median root-mean-square heat error was 0.73 $\mathrm{kJ\,mol^{-1}}$ for P, 0.70 for V-pen, 3.06 for A-hard and 9.9 for F-none; because three of the five truth families lie inside or next to P's function class and one inside A-hard's, these pooled medians reflect the benchmark design as much as the models.

Against F-none, P's heat error was smaller in every family, by factors of $R = 0.007$ to 0.056. Against A-hard it was smaller in the four families outside the characteristic-curve family ($R = 0.023$ to 0.41); on the Dubinin-Astakhov truth, which lies inside A-hard's family and whose heat is exactly the vaporisation enthalpy plus A, A-hard was far better ($R = 114$). Against the penalty model the differences were small: P was better on the strong-site family ($R = 0.66$, 0.53 to 0.84), better with an uncertain size on the two-site family with a heat capacity ($R = 0.83$, 0.74 to 0.98), and equivalent on the Dubinin-Astakhov family ($R = 0.98$); on the Fowler-Guggenheim and capacity-loss families the comparison was inconclusive, and V-pen's heat failed on three materials of the Fowler-Guggenheim family.

Descriptive and exploratory analyses

Absolute errors. Over the 26 extrapolation fits, the median error was 6.0% of q_{ref} for A-hard, 8.4% for C1, 8.5% for V-pen, 9.7% for P and 10.8% for F-none. Predicting a middle temperature is easier: P's median error in I-mid was 4.6%. In that task (descriptive and unadjusted, with code written after the verdicts) the verdict words read equivalent for P against V-pen ($R = 1.02$, 0.91 to 1.20) and to C1 ($R = 1.00$, 0.92 to 1.13) within the 25% margin, and slightly but not clearly more accurate than A-hard ($R = 0.78$, 0.62 to 1.02).

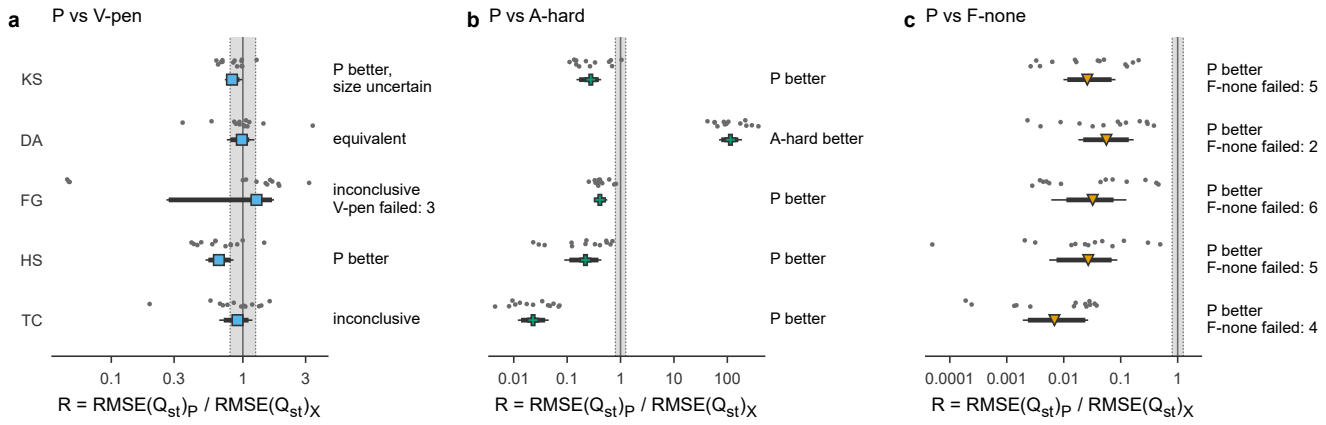


Figure 3: **Isosteric heat on synthetic truths (H2).** Ratio of the root-mean-square error of the heat at matched loading, P over the comparator (log scale; below one, P is closer to the exact heat), for each of the five truth families: Hodges-Lehmann estimate with exact 95% (thin) and 90% (thick) intervals over the 13 materials; dots, single materials (a failed task is ranked worst); shading and dotted lines, 0.8 to 1.25. KS, two-site isotherm with a heat capacity; DA, Dubinin-Astakhov; FG, Fowler-Guggenheim; HS, strong hydrophilic sites; TC, capacity falling with temperature (Note S2); KS, HS and TC lie inside or next to P's function class, DA inside A-hard's, FG inside neither. Verdict words are Holm-adjusted over the fifteen tests; "size uncertain" marks a Holm-significant difference whose size is uncertain. **a**, against V-pen; **b**, against A-hard; **c**, against F-none.

The constraints at equal structure. Against V-none, the same model without its shape constraints, P's extrapolation error ratio was $R = 1.14$ (0.94 to 1.50), and its training loss was higher by a median factor of 2.3 over the 39 fits: the constraints cost some fit to the training temperatures and bought no clear gain in extrapolation. Against the relative-pressure network F-rel, $R = 1.13$ (0.70 to 1.89). On Mg-SU-102, the one material with enough temperatures for the Kirchhoff variant, P's error was 0.74 times that of P-K. The per-material differences did not grow with the distance of the held-out temperature from the training temperatures (Spearman correlations -0.16 to 0.26 across the four comparisons).

Seed spread. The five seeds of a fit usually agreed: the median range of $\log_{10} E$ over seeds was 0.04 decades for P, 0.01 to 0.13 for the other cells with a van't Hoff or characteristic-curve structure, and 0.18 to 0.67 for the free networks (Table S4). The exceptions are informative. For P on COF-SO_3H in E-high, the five seeds reached training losses within 9% of each

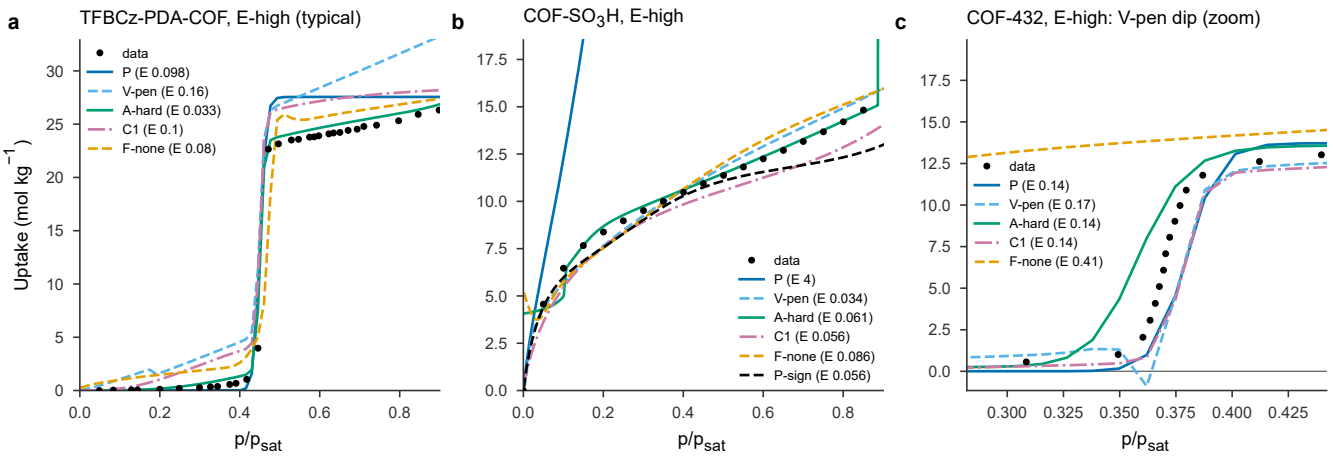


Figure 4: **Example extrapolations at the held-out temperature** (exact refits of the reported seeds; ledger X-4). Black points, the measured held-out isotherm; curves, the models' predictions, with each fit's error E in the legend. **a**, The material whose error for P in E-high is closest to P's median error in that task. **b**, COF-SO_3H in E-high, P's largest error: the reported P seed has a term with a negative heat and a capacity rising with temperature; the dashed black curve is the exploratory P-sign variant (heats positive, capacities not rising with temperature; ledger X-6). The uptake axis is clipped: P's curve leaves it and reaches 144 mol kg^{-1} at $p/p_{sat} = 0.90$, against at most 15 measured. A-hard and F-none do not pass through the origin (the characteristic curve and the free network have no Henry limit), and A-hard rises steeply near saturation, where the adsorption potential goes to zero. **c**, The V-pen fit whose held-out isotherm falls most below the highest uptake it reached at lower pressure.

other (refits, post hoc) and gave held-out errors from 0.035 to 7.7; the pre-registered rule, the lowest training loss, selected a seed with $E = 4.04$. That seed has a term with a negative heat ($Q_k = -129$ kJ mol^{-1}) and a capacity rising with temperature ($\alpha_k = -0.043$ K^{-1}), so its predicted uptake grows when the temperature is raised beyond the training range (Figure 4b). Across P's 26 extrapolation fits the rule selected the seed with the best held-out error 6 times and the worst 9 times, about what chance would give; for P, with two training temperatures, the training loss did not identify the temperature dependence.

Two exploratory checks. The following analyses were not pre-registered; they were added after the verdicts were known, because of the seed spread above (Methods). With each fit's error replaced by its median over the five seeds, for every model, the ratio against V-pen was $R = 1.00$ (0.90 to 1.24), inside the non-inferiority margin. With COF-SO_3H left

out and the selected seeds kept, it was $R = 1.05$ (0.91 to 1.35), still above the margin. The pre-registered verdict stands. Neither check removes the shortfall on its own: averaging over seeds changes the error of every fit of every model, and without COF-SO_3H the spread of the other materials still leaves the bound above the margin.

Sensitivity to the data. The H1 comparisons were repeated on the seven pre-registered dataset variants (Table S6). The ratio against V-pen stayed between 0.97 and 1.19; its verdict was inconclusive in five variants and equivalent in two, and the non-inferiority bound fell below 1.25 in one: without the coverage rule, which admits 23 materials including isotherms with occluded markers, the ratio was $R = 1.05$ (0.97 to 1.20). In the same variant P extrapolated better than F-none ($R = 0.59$, Holm-adjusted $p = 0.012$), and the characteristic curve came close to being better than P ($R = 1.52$, 1.06 to 2.19, Holm-adjusted $p = 0.064$). Shifting every digitised isotherm by the largest measured systematic digitisation error changed no verdict except that against F-none in one of the two perturbed copies, where it read "P extrapolates better".

One calorimetric value. For MOF-801, the one material with multi-temperature isotherms and a calorimetric water heat (an integral value of 53.3 $\mathrm{kJ\,mol^{-1}}$ at 30 °C[35]), every model fitted to its five temperatures gave a heat between 48.5 and 54.1 $\mathrm{kJ\,mol^{-1}}$ in the loading window of the measurement (P, 51.0). One integral value cannot rank the models; moreover the isotherms (from a different study and sample than the calorimetric value) have seven points per temperature, which fails the inclusion rule, and V-pen had only 9 states in the loading window.

Where the step lands. For water in these frameworks the uptake step is the quantity that moves with temperature. This analysis was declared in advance, but its script and the reading of its definition date from after the verdicts (descriptive). Refitting the reported seed of every extrapolation fit of the six models in the H1 comparisons (156 fits, each reproducing

its recorded error exactly), the median absolute error in the relative pressure of the step at the held-out temperature was 0.011 for V-pen, 0.013 for C1 and 0.014 for both A-hard and P (0.014), against 0.028 for F-none and 0.059 for F-hard. The structured models had similar median step errors; the free temperature networks had larger ones.

Discussion

What the shape constraints buy. That P, F-hard, A-hard and C1 never produced a decreasing or negative isotherm follows from their construction (positive capacities in a sum of increasing terms); the outcome of the test is the other half: the penalty model V-pen, at the weight chosen for accuracy on the development materials ($\lambda = 0.001$), left shape violations in 9 of 26 fits, as many as the same function with no penalty (8). As pre-registered, this is a statement about V-pen at that weight, and the post hoc refit suggests that it does not extend to heavier weights: weighted 10^6 times more, the monotonicity penalty removed every violation the count detects without a clear change in the error ratio against P. On these data a heavily weighted penalty was therefore not detectably less accurate than the hard constraints and, to the tolerance of the count, as monotone; the accuracy comparison itself stays inconclusive, with a cost of up to 65% not excluded. What separates the two is how the property is obtained. The penalty needed a far heavier weight than the one chosen for accuracy, a second choice that the joint weight of the pre-registered design did not make; the weight that worked was found on the test materials themselves, so whether it transfers to new materials is untested. The penalty also acts only at its collocation points (the unlabelled points where it is evaluated): at the heaviest weight 5 fits still had decreasing or negative points below the tolerance, 2 of them where the penalty was exactly zero. A hard constraint makes the property independent of these choices; here it raised the training loss by a median factor of 2.3 over the unconstrained function without a clear change in extrapolation error. For a process model that integrates an isotherm over a temperature swing, a non-monotone

stretch can make an adsorber appear to release water on loading, and such errors are found only if someone looks for them.

What they do not buy. The shape constraints say nothing about how the isotherm moves with temperature. P's heats Q_k and expansion coefficients α_k were free in sign, and in 14 of 26 extrapolations P's isosteric heat left the range 0 to 150 kJ mol^{-1} somewhere on the grid, 10 times with negative values. On COF-SO_3H, two training temperatures did not fix the sign of the temperature dependence: P's five seeds fitted them equally well, and the selection rule picked one whose largest-error term has a negative heat and a capacity rising with temperature. V-pen, V-none and C1 have the same sign freedom; on that fit their reported seeds happened to extrapolate well (V-pen, $E = 0.034$). For water, whose adsorption is exothermic, $Q_k > 0$ and $\alpha_k \geq 0$ are natural constraints that P lacks; with both, every term's affinity and capacity fall with temperature, so the heat is positive by construction. In an exploratory refit with these two constraints added and nothing else changed (P-sign; same initial function per seed; post hoc, Methods), the COF-SO_3H extrapolation fell from $E = 4.04$ to 0.056, with all five seeds between 0.056 and 0.17. P-sign was not better everywhere: on COF-482-hydrazide in E-high its error was 0.60 against 0.093 for P, and 8 of its 26 extrapolations still had heats above 150 kJ mol^{-1} somewhere on the grid. Its error ratios were $R = 0.81$ (0.54 to 1.05) against P, 0.96 (0.77 to 1.16) against V-pen, 0.98 (0.81 to 1.09) against C1 and 1.16 (0.70 to 1.99) against A-hard. In 131 of its 156 expansion coefficients the fit sat at the bound, $\alpha_k \approx 0$, so the variant behaved mostly as P with positive heats and no expansion term. It was designed after the confirmatory results were known, from one failure, and is a hypothesis for a future pre-registered test, not a result of this one.

What the accuracy test can and cannot say. The test declared primary did not succeed: the upper bound of the error ratio against V-pen, 1.65, exceeded the 1.25 margin, so the data do not exclude that the hard constraints cost more than 25% in extrapolation accuracy. They do not indicate such a cost either: the point estimate is $R = 1.10$ and P

was more accurate on 6 of the 13 materials. The criterion had a probability of only 0.51 to 0.74 of being met under equal accuracy even at the planned spread, which is why the pre-registration made the estimates with their intervals the primary result and the verdict words a secondary layer. The observed spread of the per-material differences, 0.30 decades, comes almost entirely from COF-SO_3H (without it, 0.13 decades, below the planned 0.158); without that material the estimate is slightly unfavourable to P and the bound, with twelve materials, stays above the margin ($R = 1.05$, up to 1.35). With errors averaged over seeds the bound is 1.24. We report the pre-registered verdict and do not claim non-inferiority. The pre-registered prediction that P would extrapolate better than the generic network F-none (H1d) was not confirmed either ($R = 0.63$, 0.40 to 1.22).

Temperature structure and the number of temperatures. The descriptive contrasts suggest, without testing it, that the temperature structure limits extrapolation more than the shape constraints do. The free temperature networks had the largest seed spread, the largest errors and the largest median step errors, and P extrapolated better than the free network with the same shape constraints on 11 of 13 materials, including both materials with three or more training temperatures, although the synthetic worlds show that this comparison cannot fail with two. Among the models fitted to every material, the characteristic curves had the lowest median errors (5.9% of q_{ref} with a penalty and 6.0% hard, against 9.7% for P), and the estimated advantage of the hard characteristic curve over P was larger on the COFs ($R = 2.67$) than on the MOFs (0.69); these are estimates without intervals at 8 and 5 materials, and class is confounded with source paper. The overall comparison was inconclusive ($R = 1.51$, 0.84 to 2.87). The characteristic curve is the standard temperature model for water in MOFs,[9] and these results give no reason to prefer a van't Hoff mixture over it for water in these frameworks. Eleven of the 13 materials offer only two training temperatures in the extrapolation tasks; with two temperatures, fits of equal training quality disagreed by two orders of magnitude at a third, as P's seeds did on COF-SO_3H. The number

of temperatures was not varied in this design, so its effect is an interpretation, not a result; a test of it needs materials measured at many temperatures, such as Mg-SU-102. Until then, learned isotherm models fitted to so few temperatures should report the spread over restarts, and water isotherms measured at four or more temperatures, with deposited numbers, would probably do more for temperature extrapolation than any constraint tested here.

The heat follows the isotherm. A model's isosteric heat is a temperature derivative of its isotherm, so it inherits the weaknesses of the temperature structure. On synthetic truths with exact heats, three of which lie inside or next to P's function class, the unconstrained network gave no defined heat in 28 of 130 tasks, and where both models gave one its error was 7 to 42 times larger than P's; the characteristic curve was exact on its own family and several times worse elsewhere, because it fixes the heat to the vaporisation enthalpy plus the adsorption potential. P and the penalty model gave heats of similar quality (median errors 0.73 and 0.70 $\mathrm{kJ\,mol^{-1}}$), with P ahead on the families with a strong-site or heat-capacity contribution. P's median heat error per family was 0.09 to 1.45 $\mathrm{kJ\,mol^{-1}}$, including the two families outside its function class. None of this measures the heat of a real material, for which calorimetric data are missing, and within the penalty domain around the measured data P's heat left the physical range in 13 of its 26 extrapolations.

Recommendations for learned isotherm models. Several practices follow from this study for anyone who builds or evaluates a learned isotherm model with few temperatures. Count shape violations on a grid finer than the points where any penalty is evaluated, and report the count: a penalty weight chosen for accuracy may leave violations in a third of the fits, and a weight that removes them has to be chosen for that purpose, and checked on materials not used to choose it. Where a guarantee is wanted without that second tuning, build the property into the model; here no accuracy cost was detected, although one of up to 65% could not be excluded. Constrain the temperature dependence as well as the shape, at least the sign of the heats; the shape constraints alone allowed P's worst failure. Report the

spread of the held-out error over restarts, which with two training temperatures can reach two orders of magnitude (Table S4). Compare with the characteristic curve, which had the lowest median error here, and fix the comparison, its margin and the words of each outcome before the confirmatory fits, with synthetic data on which each verdict can be checked to be able to fail (Note S1).

Limitations. The confirmatory set has 13 materials, the number of MOFs and COFs with water isotherms at three or more temperatures that passed our inclusion rules, and most of the isotherms were digitised from published figures. The digitisation error was measured (Note S3); it sets a floor of about 0.003 on E, well below the smallest observed errors (0.008 or more). Occlusion of markers excluded several data sets, and the excluded materials may differ from the included ones; because occlusion happens where isotherms coincide in relative pressure, close to where the characteristic curve holds, the exclusion may have removed materials favourable to A-hard, which remains a caveat on that comparison. The materials are not fully independent: three COFs come from one paper and two from another, and averaging the per-material differences within each source paper left the primary comparison unchanged in direction ($R = 1.12$, 0.92 to 3.08 over 10 sources; Table S7). The isotherms of two materials start only at relative pressures of 0.08 and 0.39. Each material's temperatures span 15 to 40 K, and the held-out temperature lies 5 to 25 K (median 10 K) beyond the nearest training temperature, mostly from two training temperatures; these steps are much shorter than the swings of the adsorption cycles that motivate the study. With two training temperatures, P's 18 parameters include a heat and an expansion coefficient for each of its four terms, which two isotherms cannot identify; this is the mechanism behind the seed spread on COF-SO_3H. The hyperparameters were tuned on development materials, one of which comes from the same paper as three confirmatory COFs, so a mild transfer of tuning to those materials cannot be excluded. The models describe one adsorption branch; hysteresis and framework flexibility are outside their scope. No confirmatory material has a

calorimetric water heat, so no claim is made that any model's heat is closer to the true heat of a real material. The Henry limit cannot be tested on these data, because no isotherm reaches the Henry regime, so the finite Henry slope of P is a property of the model, not a validated prediction.

Conclusions

Building monotonicity and a Henry limit into a potential-based isotherm model gave monotone, non-negative extrapolations in every fit, which training penalties at the weight chosen for accuracy did not, at an accuracy cost that this pre-registered test could neither exclude beyond the 25% margin nor detect. In a post hoc refit, a far heavier penalty weight gave the same shape to the tolerance of the count; the hard constraint gives the guarantee without that second choice, and no accuracy advantage was found. The same constraints left the temperature dependence free, and the model's worst extrapolation and its out-of-range heats came from that freedom. For temperature extrapolation of water isotherms in these frameworks, the descriptive evidence points to the temperature structure, including the signs of the heats and expansion coefficients, as what to constrain next, and to more measured temperatures as what to measure.

Methods

Pre-registration

The design, the data, the models, the tuning rules, the metrics, the statistical tests and the words of every verdict were fixed in a pre-registration and frozen before any confirmatory fit. The freeze is a public, tagged commit (`prereg-p3a-freeze`) whose manifest holds the SHA-256 of 71 files (the text, the data, the tuned configurations and the fits behind them, the synthetic-world and power results, and every script that builds, fits or scores, including

a second scorer written separately from the frozen text by an AI agent, Claude, without importing the first's code) and the package versions; the confirmatory scripts refuse to run if any of them differs. Every defect found in the design, code or data, before or after the freeze, is listed with its cause and fix in a public ledger (25 entries before the freeze and 7 after the verdicts).

Data and curation

Water adsorption isotherms of MOFs and COFs measured at several temperatures on one sample were collected from the primary literature and from the NIST/ARPA-E adsorption database.[36] The confirmatory data come from published studies,[2,8,37–44] and the development data from two further studies and one of the same;[42,45,46] Table S1 gives the source of each material and Figure S1 shows the data. A material entered the confirmatory set if, at three or more temperatures, its adsorption isotherm had at least 8 points with $0 < p/p_{\mathrm{sat}} \leq 0.90$, its figure passed the digitisation quality gates, its temperatures were unambiguous, and its curve was covered: no gap above 0.25 in p/p_{sat}, and a first point at $p/p_{\mathrm{sat}} \leq 0.10$ or at an uptake of at most 0.10 q_{ref}. Where a paper gave both measured isotherms and curves derived from them, the measured isotherms were used. The confirmatory set has 13 materials, 5 MOFs and 8 COFs; every decision and its reason is in the inclusion log. Four further materials form a development set used only to choose hyperparameters.

Isotherms published only as figures were digitised before any model was fitted. Vector figures were read from the PDF drawing commands, with marker centres mapped through the axis ticks; raster figures by template matching with the figure's own marker footprint, with tick marks located to sub-pixel precision and an open-marker veto against desorption points. No physical prior was used to filter points. The method was validated against published numeric data. On Mg-SU-102 the vector extraction recovered 198 of 198 points with a root-mean-square uptake error of 7×10^{-6} g g^{-1}. On raster figures of HKUST-1 the root-mean-square uptake error was 0.11 to 0.19% of the axis span, or 0.0012 to 0.0034 of the

isotherm's maximum, with no false detections.

Only adsorption branches with $p/p_{\mathrm{sat}} \leq 0.90$ were used. Relative pressures were converted with the IAPWS saturation pressure, absolute pressures were used as reported, and uptakes were converted to $\mathrm{mol\,kg^{-1}}$. The scale q_{ref} of a material, its largest uptake in this range over all temperatures, is used only for scoring.

Models

Every model is a function $q(s, T)$ of $s = \ln(p/\mathrm{Pa})$ and temperature, and the isosteric heat of every model is computed by the same code, $Q_{\mathrm{st}} = -RT^2(\partial q/\partial T)/(\partial q/\partial s)$, by automatic differentiation.

The potential model (P) is fitted as a reduced spreading pressure whose derivative is the uptake:

$$\psi(s,T) = \sum_k \frac{c_k(T)}{g_k}\,\mathrm{softplus}\Big(g_k\,(s-\theta_k(T))\Big), \qquad q = \frac{\partial \psi}{\partial s} = \sum_k c_k(T)\,\sigma\Big(g_k\,(s-\theta_k(T))\Big), \quad (1)$$

with σ the logistic function and $\mathrm{softplus}(x) = \ln(1 + e^x)$. Each term is a Langmuir term ($g_k = 1$, the Henry terms) or a cooperative Sips-type term ($g_k \geq 1.5$) in $K_k(T)\,p$ with $\ln K_k = -\theta_k$, so P is a positive mixture of Langmuir and cooperative terms with van't Hoff affinities, $\theta_k(T) = \theta_{k0} - (Q_k/R)(1/T - 1/T_{\mathrm{ref}})$, and capacities $c_k(T) = c_{k0}\exp[-\alpha_k(T - T_{\mathrm{ref}})]$ with $c_{k0} > 0$. The tuned P has two Henry and two cooperative terms and 18 parameters. The capacities and the exponents are constrained; the heats Q_k and the expansion coefficients α_k are not, in sign or size. For every parameter value the uptake increases strictly with p, q/p tends to a positive Henry constant as $p \to 0$, and the heat is finite, because $\partial q/\partial s > 0$; the sign and size of the heat are not constrained. C1 is a two-term member of the same broad family with free exponents. Every model's equations and tuned settings are in Note S4 and Table S8.

The models form a factorial of shape constraint (none, penalty, hard) and temperature

structure: a free network in $1/T$ (F), van't Hoff per term (V), or the temperature-invariant characteristic curve in A (A). The nine cells are F-none, F-pen and F-hard; V-none, V-pen and P (the hard van't Hoff cell); A-none, A-pen and A-hard. F-none and F-pen are fully connected networks of $(s, 1/T)$; F-hard has P's potential and shape constraints with capacities and positions given by a network of $1/T$. In the V and A columns, the unconstrained and penalised models have the function of P or of A-hard without the constraints, the same initial function and the same tuned structure; the penalty models add a monotonicity penalty and (except for the characteristic curve) a scale-free Henry-limit penalty on collocation points within 15 K of the training temperatures, which include the held-out temperature in all but one of the 26 extrapolations. A dual-site Sips model with van't Hoff affinities (C1) is the classical comparator. Two further models are reported descriptively: F-rel, a network of $(\ln p/p_{\mathrm{sat}}, 1/T)$, and P-K, P with a heat capacity per term.

Fitting and tuning

Each model was fitted by Adam followed by L-BFGS on the mean squared error of q scaled by the largest training uptake, in double precision, from 5 seeds; the fit with the lowest training loss was kept, never selected on held-out error. Every scale of a fit came from the training temperatures only. Hyperparameters were chosen on the development set alone, in two stages: the models that own a structure tried up to 12 configurations each, and the unconstrained and penalised models inherited the chosen structure, the penalty models tuning only the penalty weight over 10^{-4} to 100.

Tasks and outcomes

For a material with temperatures $T_1 < \cdots < T_k$, the task E-high trains on $T_1, \ldots, T_{k-1}$ and predicts T_k; E-low trains on $T_2, \ldots, T_k$ and predicts T_1; the interpolation task I-mid holds out a middle temperature ($T_{\lceil k/2 \rceil}$) and trains on the rest. The primary outcome is the held-out error $E = \mathrm{RMSE}(q_{\mathrm{pred}} - q_{\mathrm{obs}})/q_{\mathrm{ref}}$, summarised per material as the mean of $\log_{10} E$

over the two tasks. Physical violations were counted on a dense grid: a fit violated if an isotherm fell by more than 2% of q_{ref} below its running maximum, or went below -2% of q_{ref}, inside the domain where the penalties act; violations were counted over the fits of the two extrapolation tasks. Because no material here has a calorimetric water heat, the heat was tested on synthetic isotherms with exact heats, built on every material's own temperatures and pressures from five truth families, with noise of 0.4% of q_{ref}.

Isosteric heat on synthetic truths

The five truth families are a Langmuir site plus a cooperative Sips site with a heat capacity, a Dubinin-Astakhov curve, a Fowler-Guggenheim isotherm with lateral attraction, strong hydrophilic sites with a heat that falls steeply with loading, and a Sips isotherm whose capacity falls with temperature (Note S2). Each family is sampled on every confirmatory material's own temperatures and pressure grid, with shape parameters fitted to that material's middle-temperature isotherm and the thermodynamic parameters fixed in advance. P, V-pen, A-hard and F-none were fitted to the E-high and E-low tasks of every truth. The primary metric compares the heats at equal loading: at each held-out state where the model's own loading lies between 0.1 and 0.9 q_{ref}, the model's Q_{st} is compared with the truth's Q_{st} at the same loading and temperature, found by inverting the monotone truth, and the root-mean-square difference is taken in $\mathrm{kJ\,mol^{-1}}$. A task fails, and is ranked worst, if any compared heat is undefined or not finite, if fewer than 3 states fall in the window, or if the truth cannot reach the model's loading. The fifteen comparisons (three models against P in five families) are Holm-corrected as one family. The single calorimetric value found for a material with multi-temperature isotherms, MOF-801,[35] is used only in a descriptive check; no claim about the heat of a real material is made.

Admissibility

For each model the violating fits are counted over the 26 fits of the two extrapolation tasks, with a Wilson 95% interval; the counts over all tasks, adding the interpolation task, are reported beside them. The words follow a table fixed in advance: at least 20% of fits when the lower bound reaches 20%, no violations detected when no fit violates, fewer than 20% when the upper bound is below it, and an uncertain rate otherwise. The pre-registered prediction is that penalties leave violations that the hard constraints remove: V-pen has at least one violating fit and P has none. It concerns V-pen at the penalty weight chosen for accuracy on the development set ($\lambda = 0.001$), not penalties at every weight. By construction P cannot violate; a violating fit of P would be a bug and halts the analysis.

Descriptive and sensitivity analyses

Fixed in advance and reported without a hypothesis: the interpolation task; P against V-none (the pure effect of the constraints at equal structure), against F-hard (rigid against free temperature dependence), against the multilayer perceptron and against the Kirchhoff variant of P; MOFs and COFs separately; the error against the distance of the held-out temperature from the nearest training temperature; and the spread of the characteristic curve across temperatures. The H1 comparisons were repeated on the datasets given by earlier versions of the coverage rule and by a 7-point rule (6, 9, 23 and 14 materials), on Co_2Cl_2(BTDD) from its characteristic curves, and on two perturbed copies of the data in which every digitised isotherm is shifted in p/p_{sat} by ± 0.002, the largest systematic shift measured in the digitisation benchmark.

Analyses added after the verdicts

The following were specified only after the confirmatory verdicts were known and are listed in the correction ledger (entries X-1 to X-7): the form of the seed-spread report and two

exploratory checks of the primary comparison (X-1); the code of the declared interpolation comparisons and of the step-position error (X-2); a wrapper that feeds the frozen sensitivity script the sharded run files (X-3); refits of the reported seeds to split the heat-range count by sign and domain, record P's fitted heats and save the predicted isotherms, the class split of every comparison, a same-source check and the heat comparison without imputed failures (X-4); the reporting of the pre-registered heat-range count, omitted from the first draft (X-5); an exploratory variant of P with positive heats and non-negative expansion coefficients, P-sign, designed after a failure seen in X-4 and defined in Note S4 (X-6); and the refit of V-pen with the monotonicity penalty on its own, heavier weight, the heat-range count restricted to the penalty domain and the distance of each extrapolation from the nearest training temperature, added at an internal review (X-7). The pre-registration requires the seed spread of the held-out error to be reported without fixing its form; we report, per model, the median over its extrapolation fits of the range of $\log_{10} E$ over the 5 seeds, and how often the seed with the lowest training loss had the best or the worst held-out error. Two exploratory checks of the primary comparison followed from it: every fit's error replaced by its median over the seeds, for every model, which uses no held-out information to choose among seeds; and the material with the largest paired difference left out. Both reuse the frozen analysis code. The step-position error, declared in advance but without a frozen script, needs the fitted curves, which the run records do not keep: the selected seed of each extrapolation fit of the models in the confirmatory comparisons was refitted with the frozen code and configuration, and the refit had to reproduce the recorded error. The step position of a curve at the held-out temperature is the p/p_{sat} at which it first reaches half of its own uptake at $p/p_{\mathrm{sat}} = 0.90$.

Software

All fits ran in double precision on a CPU with Python 3.12.10, PyTorch 2.12.0, NumPy 2.4.6, SciPy 1.17.1 and pandas 3.0.3; the versions are part of the frozen manifest.

Statistics

The unit is the material ($n = 13$). For each comparison of P with a model X, $d_m = e_m(\mathrm{P}) - e_m(\mathrm{X})$ was tested with the exact two-sided Wilcoxon signed-rank test, with Holm correction over the four confirmatory comparisons (penalty model, characteristic curve, generic network, dual-site Sips). The effect is the Hodges-Lehmann estimate with exact distribution-free 95% and 90% intervals, reported as the error ratio $R = 10^d$; these intervals assume a symmetric distribution of d_m, and percentile bootstrap intervals, which do not, are given in Table S7. A failed fit was ranked worst, never dropped. The words of each verdict follow a table fixed in advance, with a smallest effect of interest of 25% ($R = 0.8$ or 1.25). The one declared primary claim, that the hard constraints cost no more than 25% in accuracy against the penalty model, was judged by non-inferiority: the upper exact 95% bound of R below 1.25 (one-sided 2.5%), with the 98.75% bound reported beside it. At 13 materials its exact size at the margin is 0.024 (0.005 for the 98.75% bound). Before the freeze, the power of each verdict was simulated with the analysis code itself, and the whole pipeline was run on five synthetic worlds, with an expected outcome written for each before it ran, to show which verdicts can fail (Note S1). Three of the expectations were missed and are reported as missed, and the comparison of a free with a van't Hoff temperature dependence was made descriptive because it could not fail.

Data and Software Availability

The pre-registration, the freeze manifest with the SHA-256 of every frozen file, the corrections ledger, the digitisation provenance of every source (calibration, gates and per-point records), all run records and all results are public at `https://github.com/abdulhamidbatayhi123/ai-isotherm-thermo` and archived with a DOI on Zenodo.[47] The isotherm data themselves are third-party data: they were digitised from figures published by others or taken from records of the NIST/ARPA-E adsorption database,[36] and the digitised and database point

lists of the 13 confirmatory and four development materials are not redistributed, because the figures remain under their publishers' copyright and licences. The manuscript remains fully checkable without them: every source is cited (Table S1), the digitisation scripts regenerate the point lists from the published figures (and from the database), the per-point provenance and the calibration of every digitisation are public, and the SHA-256 of the data set used is in the freeze manifest, so a regenerated set can be checked against it exactly. The point lists are available to the editor and reviewers in confidence on request. All code (Python, MIT licence) is in the same repository and archive, with the exact package versions in requirements-lock.txt; the manuscript build checks every number in the text against its source file.

Supporting Information Available

Data set, per-material error ratios, violating fits, seed spread, isosteric-heat tests, dataset variants, bootstrap intervals and model settings, and the penalty-weight refit (Tables S1–S9); confirmatory isotherms (Figure S1); synthetic worlds and power, synthetic truths for the heat, digitisation error and model definitions (Notes S1–S4) (PDF).

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